CIRCUITBRIDGE ELECTRONICS

Business profile & technical sales assistant brief | Simulated online retailer | EECE 503 agent project

Fictional demo notice. All products, prices, named team members and policies below are created for an academic assignment. No real company, inventory, checkout, delivery or customer communication exists.

Our mission

Make engineering hardware accessible without sacrificing technical accuracy. We help students, makers, university laboratories and small product teams evaluate electronics products, understand specifications and choose suitable equipment.

Our unique value proposition

An engineering-first buying experience: **clear specifications, budget-aware recommendations, straightforward USD demo prices, and itemized non-binding quotations**. NOVA explains trade-offs before trying to collect a contact lead.

Services

Technical product guidance: Compare microcontrollers, power tools, sensors, lab instruments, networking devices and accessories against known specifications.

Product recommendations: Suggest suitable products from a fixed demo catalog, considering needs and stated budget.

Non-binding quotations: Multiply published catalog unit prices by requested quantities. Tax, shipping and customs are excluded.

Lead collection: Optionally capture a customer name, email and short technical inquiry for a demonstration follow-up workflow.

Feedback and unanswered questions: Log gaps in the catalog/business knowledge so a human administrator could review them.

Meet the fictional team

Dana Rahim - Founder & Technical Director. Background in embedded systems and product validation.

Elias Haddad - Product Engineer. Focus on RF/networking equipment and laboratory instrumentation.

Noor Karam - Customer Experience Lead. Designs clear support and lead-follow-up workflows.

Who is NOVA?

NOVA is the fictional store's AI technical sales engineer. It answers questions using this profile, a short text summary and a structured JSON product catalog. NOVA must never invent a price or business policy; unknown questions are logged for review. It is not a human representative.

DEMO PRODUCT CATALOG

Authoritative item identifiers and demonstration unit prices in USD. The machine-readable product_catalog.json contains full specs and search keywords.

SKU	Product	Category	USD
CB-ESP32-S3	ESP32-S3 Development Board	Microcontrollers	$18.00
CB-RP2040	RP2040 Maker Board	Microcontrollers	$12.50
CB-BREAD-KIT	Breadboard Prototyping Kit	Prototyping	$24.00
CB-SENSOR-10	10-in-1 Sensor Pack	Prototyping	$35.00
CB-DMM-6000	6000-Count Digital Multimeter	Lab Equipment	$49.00
CB-LOGIC-8	8-Channel USB Logic Analyzer	Lab Equipment	$28.00
CB-SOLDER-75	75 W Digital Soldering Station	Lab Equipment	$89.00
CB-SCOPE-50	50 MHz Dual-Channel Oscilloscope	Lab Equipment	$245.00
CB-PD-65	65 W USB-C GaN Charger	Power	$34.00
CB-PSU-30	Adjustable 0-30 V Bench Power Supply	Power	$112.00
CB-SWITCH-5G	5-Port Gigabit Ethernet Switch	Networking	$31.00
CB-WIFI-USB	Dual-Band USB Wi-Fi Adapter	Networking	$39.00
CB-ROUTER-AX	Wi-Fi 6 Lab Router	Networking	$128.00
CB-USB-CABLE	USB-C Cable Set (3 Pack)	Accessories	$15.00

Business and quotation policies

Pricing: prices are fictional catalog unit prices in USD. A quotation includes item subtotal only; shipping, tax, customs and installation cannot be determined from the catalog. No discounts are promised.

Availability and delivery: no live stock or order tracking exists. Shipping coverage, delivery dates, payment options and warranty terms are not documented and must be treated as unknown.

Privacy: lead and feedback data are stored in local private JSONL demo logs, not transmitted to store staff. The repository must never contain API keys or collected customer information. Public hosting may reset local files.

Sample calculation: two CB-SOLDER-75 stations at $89.00 each = $178.00 item subtotal. Tax and shipping excluded. This is not a binding offer.